TTM Technologies (TTMI)

PCB Cycle Turn-around Presents a Selective Entry Point

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TTM Technologies operates at the intersection of high-mix low-volume PCB demand and emerging advanced packaging needs, a position that has been overshadowed by near-term inventory corrections in the consumer electronics sector. While consensus focuses on short-term revenue volatility, the company's diversified end-market exposure, disciplined capital allocation, and improving free cash flow conversion suggest a more durable earnings trajectory. We therefore maintain a Hold — Valuation Concerns rating with a blended target price of $76.71, implying -27.7% upside to the current reference price of $106.04.

HOLD	$76.71	-27.7%	$106.04
Hold — Valuation Concerns	BLENDED TARGET	BLENDED UPSIDE	REFERENCE PRICE
INVESTMENT RATING			

Investors are reacting to quarterly order softness as if it were a structural demand shock, yet the underlying multi-year drivers of 5G infrastructure rollout, automotive electrification, and data-center server upgrades remain intact and are poised to reaccelerate as inventory cycles normalize.

EXECUTIVE SUMMARY

TTM Technologies is a leading global manufacturer of printed circuit boards, serving a broad mix of end markets that include communications infrastructure, aerospace and defense, automotive, medical, and industrial sectors. The company's footprint spans over 30 manufacturing sites across Asia, North America, and Europe, providing it with geographic diversification and the ability to shift production in response to regional demand fluctuations. In terms of market share, TTM ranks among the top five PCB suppliers worldwide, with a particularly strong position in high-reliability, high-mix low-volume boards that command premium pricing and longer customer relationships. The recent overreaction in the stock price stems from a confluence of temporary headwinds: a slowdown in consumer-driven smartphone and PC orders, elevated raw material costs, and a cautious stance by original equipment manufacturers as they work through excess inventory built during the pandemic-era surge. These factors have compressed near-term revenue growth and pressured multiples, even though the company's backlog for longer-cycle products remains healthy and its order book for advanced packaging substrates shows sequential improvement. Capital allocation has been a key differentiator for TTM. Over the past three years the firm has returned capital to shareholders through a steady dividend and opportunistic share repurchases while simultaneously investing in capacity upgrades for high-density interconnect (HDI) and flexible circuit technologies.

The balance sheet remains robust, with net debt to EBITDA below 2.0x and a liquidity cushion exceeding $400 million, providing flexibility to pursue selective bolt-on acquisitions that could enhance its advanced packaging portfolio or expand its presence in fast-growing automotive electronics.

OPERATIONAL SNAPSHOT

$106.04	$2.48B	$150M	FCF MARGIN	EQUITY BETA	$106.04
REF. PRICE	LTM REVENUE	LTM FREE CASH FLOW	6.0%	1.28	REF. CLOSE

LTM figures are Cascadia estimates grounded in the latest filings; ref. price as of 2026-10-14.

THE MOAT IS RESILIENT AND STICKY

TTM Technologies benefits from a resilient moat rooted in the high switching costs associated with its high-mix low-volume PCB offerings. Customers in aerospace, defense, and medical devices often qualify suppliers through rigorous certification processes that can take 12 to 18 months, making it costly and risky to qualify an alternate source. This creates a sticky relationship where TTM can command price premiums and enjoy longer contract durations compared with commodity-focused PCB producers. The stickiness is further reinforced by TTM’s deep technical expertise in advanced materials and processes, such as embedded passive technology and high-frequency laminates, which are critical for next-generation 5G base stations and automotive radar systems. These capabilities are protected by a substantial patent portfolio and proprietary process know-how that rivals find difficult to replicate without significant capital investment and time. Consequently, TTM enjoys pricing power in niches where performance and reliability outweigh pure cost considerations, allowing it to maintain stable margins even during broader industry downturns.

VALUATION

Three scenarios make the valuation range explicit

Our valuation framework translates three distinct outlooks for revenue growth and free cash flow margin into implied equity values. The base case assumes a modest recovery in demand driven by automotive electrification and 5G infrastructure, paired with gradual margin improvement as operational efficiencies are realized. The bear case reflects a prolonged inventory overhang and weaker-than-expected adoption of advanced packaging, while the bull case captures an accelerated rollout of 6G research initiatives and a faster shift toward electric vehicle power electronics that lifts both volume and pricing.

SCENARIO	WEIGHT	REVENUE CAGR	TARGET FCF MARGIN	TARGET	UPSIDE
Bear	25%	-0.5%	4.0%	$52.42	-50.6%

Base	50%	2.0%	6.0%	$79.17	-25.3%
Bull	25%	4.0%	8.0%	$96.09	-9.4%

Bear $52.42

The bear outcome materializes if consumer electronics demand remains subdued for an extended period, causing a persistent decline in high-volume board orders and forcing TTM to operate at lower utilization rates. In this environment, pricing pressure intensifies as competitors excess capacity drives down ASPs, and the company's fixed cost base cannot be adjusted quickly enough, dragging FCF margin down to 4% and limiting revenue growth to a slight contraction.

Base $79.17

The base case reflects a gradual normalization of inventory levels across the supply chain, allowing TTM to capture modest growth in automotive and communications infrastructure segments. Operational initiatives such as lean manufacturing and automation lift the FCF margin to the historical average of 6%, while revenue expands at a 2% CAGR as the company benefits from higher-margin HDI and flexible circuit orders.

Bull $96.09

In the bull scenario, a faster-than-expected rollout of 5G-advanced and early 6G research projects, combined with accelerated adoption of electric vehicle power electronics, drives a surge in demand for high-frequency and high-density interconnect boards. TTM's advanced packaging capabilities enable it to win share in substrate markets, pushing revenue growth to 4% and allowing FCF margin to expand to 8% as scale efficiencies and premium pricing take hold.

VALUATION FRAMEWORK

A live discount rate, not a static hurdle

The 9.01% WACC used in our model is derived from the CAPM, incorporating a risk-free rate of 3.8%, an equity risk premium of 5.2%, and a levered beta of 1.28 that reflects TTM's exposure to cyclical industrial end markets. This discount rate implies that the market is pricing the stock for a moderate level of risk, balancing the company's cyclicality against its defensive niche in high-reliability boards. A live WACC allows us to capture changes in market sentiment and interest rate expectations as they evolve, rather than anchoring to a static hurdle rate that could misstate intrinsic value.

FCF margin normalization is the engine

Across the scenarios, the primary driver of valuation dispersion is the trajectory of the free cash flow margin. In the bear case, margin compression to 4% stems from underutilized factories and heightened competitive pricing, while the base case assumes a return to the 6% level that TTM has historically achieved through disciplined cost controls and product mix shifts. The bull case envisions margin

KEY VALUATION PARAMETERS

Revenue base year	**2026**
Forecast horizon	**10 years**
Live WACC	**9.01% (2026-10-14)**
Terminal growth	**2.5%**
Equity beta	**1.28**
Diluted shares	**71.0M**

HOLD

expansion to 8% as the company leverages its advanced technology platform to capture premium-priced opportunities in emerging sectors, thereby converting incremental revenue into higher cash generation.

Mechanics are explicit

Revenue becomes EBIT, cash taxes come out, then we add back D&A and stock compensation, subtract CapEx and the change in net working capital to reach unlevered free cash flow, which we discount and divide by diluted shares.

Methodology. Detailed EBIT-bridge DCF on three scenarios, probability-weighted 25/50/25. Beta decays toward a mature peer over the forecast. Terminal value via Gordon growth. Targets are model-computed, not estimated.

over $76.71

blended target

BUSINESS & SEGMENT DRIVERS

Revenue prospects anchor on automotive electrification and 5G infrastructure HOLD

TTM's automotive segment, which supplies rigid and flexible PCBs for power electronics, ADAS sensors, and electric vehicle charging systems, has exhibited a compound annual growth rate of roughly 7% over the last three years, outpacing the broader PCB market. The company's investment in high-temperature laminates and copper-in-polyimide flex circuits positions it to capture a larger share of the EV power train market, where voltage levels and thermal management demands drive demand for more sophisticated board designs. In the communications infrastructure segment, TTM provides high-frequency laminates and backplane boards for 5G base stations, network switches, and data-center servers. The rollout of 5G-advanced (Release 16/17) and the early stages of 6G research are expected to increase the average content per node, boosting both volume and ASP for TTM's high-mix low-volume offerings. Management has indicated that the communications segment could contribute mid-single-digit percentage points to overall revenue growth by 2028, assuming a steady cadence of network upgrades. The aerospace and defense segment remains a steady contributor, characterized by long-life programs and stringent qualification requirements. While growth in this segment is modest, its high margin profile and resistance to cyclical swings provide a stabilizing effect on overall earnings. TTM's expertise in ruggedized boards and its ability to meet Mil-Std-810 standards have secured it a preferential supplier status with several prime contractors. **What to monitor**

- **Order book trends** Month-over-month changes in the backlog for high-density interconnect and flexible circuits will signal whether the inventory correction is easing or persisting.
- **CapEx efficiency** The ratio of capital expenditures to depreciation and amortization indicates whether the company is over-investing in capacity or aligning spend with actual demand signals.
- **Raw material cost pass-through** The ability to offset copper and resin price increases through price escalation clauses in customer contracts will directly affect gross margin stability.

- **Advanced packaging wins** Success in securing substrate orders for emerging chip-let and fan-out wafer-level packaging projects will be a leading indicator of bull-case upside.

$2.48B	6.0%	1.28	71.0M	$150M	2.5%
LTM REVENUE	FCF MARGIN	EQUITY BETA	DILUTED SHARES	LTM FREE CASH FLOW	TERMINAL GROWTH

METRIC	LATEST FY
Revenue	$2.48B
Gross Profit	$420M
EBIT	$180M
Net Income	$115M
Free Cash Flow	$150M
R&D	$30M
ROE	9.2%

RISK & MOAT MATRIX

The bear case

SHARE EROSION

TTM could lose market share if competitors accelerate capacity expansions in low-cost regions, particularly if the company's utilization rates remain subdued and it is

M&A DIGESTION

Integrating acquisitions in the advanced packaging space may prove challenging, potentially leading to cultural clashes, systems integration delays, and unexpected cost

REGULATORY AND PRICING SCRUTINY

Increased governmental focus on supply chain security for critical electronics could result in new localization requirements, forcing TTM to duplicate capacity or face restrictions on certain end markets.

TECHNOLOGY DISRUPTION

The rise of additive manufacturing techniques for PCBs or the adoption of alternative interconnect technologies could erode demand for traditional board fabrication, especially in prototyping and low-volume segments.

REINVESTMENT TREADMILL

To stay competitive, TTM must continually invest in costly equipment upgrades for HDI and flexible circuits; if cash flow generation falters, the firm may be forced to defer investments, weakening its technological edge.

WACC AND DURATION SENSITIVITY

Given the relatively long duration of TTM's cash flow profile, a sustained rise in long-term interest rates would increase the discount rate and materially depress the present value of future cash flows.

unable to match aggressive pricing.

overruns that dilute anticipated synergies.

The strengths

OLIGOPOLY POSITION

TTM operates in a concentrated global PCB market where the top five suppliers control over 55% of capacity, limiting the intensity of price wars and providing a degree of pricing discipline.

FCF COMPOUNDING

The company's ability to convert a stable base of high-margin orders into free cash flow enables reinvestment in growth initiatives and shareholder returns without overreliance on external financing.

GEOGRAPHIC DURABILITY

With manufacturing sites spread across Asia, North America, and Europe, TTM can mitigate regional disruptions and shift production to meet localized demand, enhancing supply chain resilience for its customers.

PIPELINE OPTIONALITY

TTM's ongoing R&D efforts in embedded passives, high-frequency laminates, and advanced packaging substrates create a pipeline of potential new products that can be commercialized as market needs evolve.

Compelling risk-reward, skewed to the upside

While near-term headwinds have created a valuation gap, the underlying fundamentals of TTM Technologies suggest that the risk-reward profile is asymmetric to the upside. The company's entrenched position in high-reliability, high-mix low-volume boards provides a cushion against severe downturns, and its strategic investments in automotive electrification and 5G-advanced infrastructure are aligned with multi-year growth trends that are only beginning to materialize. Should the inventory correction prove transitory and demand re-accelerate, the base case free cash flow margin of 6% could be sustained, supporting a fair value near $79 per share. Even in a more adverse scenario, the firm's strong balance sheet and diversified end-market exposure limit downside risk, while the bull case captures the potential for margin expansion to 8% should advanced packaging adoption outpace expectations. We therefore maintain a Hold — Valuation Concerns rating, with a blended target price of $76.71 and an implied -27.7% upside to the reference price of $106.04.

Justin Wen, Senior Equity Research Analyst, Cascadia Analytics

FRANCHISE DEEP DIVE

High Frequency Laminates repriced on normalization

TTM's high frequency laminate product line serves as a critical enabler for 5G base station antennas, military radar systems, and high-speed data-backplane applications. Historically, this line has commanded premium ASPs due to its low dielectric loss and tight thickness tolerances, characteristics that are difficult to replicate with standard FR-4 materials. During the recent inventory correction, order volumes for these laminates softened as original equipment

manufacturers deferred non-essential upgrades, leading to a temporary dip in utilization and a modest decline in ASPs as excess supply weighed on spot pricing. The growth narrative for high frequency laminates has disappointed primarily because the expected acceleration in 5G rollout has been slower than anticipated, particularly in regions where permitting and site acquisition have lagged. Additionally, the shift toward integrated antenna-module designs has reduced the laminate content per unit in some applications, offsetting volume gains. Furthermore, raw material price volatility for specialized resins and copper foil has pressured margins, limiting the ability to fully pass through cost increases to customers. **Why the growth numbers disappointed**

- Slower-than-expected 5G infrastructure deployment in key markets.
- Design-level shifts reducing laminate content per antenna module.
- Persistent raw material cost inflation for high-purity copper and specialty resins.
- Competitive pressure from alternative low-loss materials entering the market.

Our read: Normalization, not share loss or thesis break. As inventory levels normalize and 5G-advanced projects move from planning to execution, demand for high frequency laminates is expected to recover, allowing TTM to reclaim its historical premium pricing and restore margin expansion.